D3.1 Reproduction in plants

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 **STATE** the function of the stigma. [D3.1.9] [1]

2 **STATE** what a pollen tube delivers, and where. [D3.1.8] [2]

3 **DEFINE** cross-pollination. [D3.1.10] [1]

4 **STATE** two external conditions needed for germination. [D3.1.12] [2]

5 **STATE** the plant growth substance produced by the embryo during germination. [D3.1.12] [1]

Section B: Understanding

15 MARKS

6 **DISTINGUISH** between pollination and fertilization. [D3.1.8] [2]

7 **DRAW** a half-flower of an insect-pollinated plant, with five parts labelled. [D3.1.9] [5]

8 **OUTLINE** two methods that promote cross-pollination. [D3.1.10] [2]

9 **EXPLAIN** how self-incompatibility maintains genetic variation in a species. [D3.1.11] [3]

10 **EXPLAIN** why it is an advantage for a seed to be dispersed away from its parent. [D3.1.12] [3]

Section C: Working with data

14 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [D3.1.12]

Five dishes of 50 cress seeds were kept for 5 days under different conditions. The number that germinated was counted.

DISH	CONDITIONS	GERMINATED	GERMINATION / %
A	Moist, 22 °C, air	46	
B	Dry, 22 °C, air	0	
C	Moist, 4 °C, air	3	
D	Moist, 22 °C, no oxygen	1	
E	Moist, 22 °C, air, dark	45	

a **CALCULATE** the percentage germination for dishes A and C. [2]

b **EXPLAIN** the result for dish C. [2]

c **EXPLAIN** the result for dish D. [2]

d **DEDUCE** what dishes A and E show about the need for light. [1]

e **SUGGEST** why germination in dish A did not reach 100%. [1]

12 [D3.1.12]

The dry mass of 10 pea seedlings, grown in the dark, was measured over 10 days.

DAY	0	2	4	6	8	10
DRY MASS / G	2.50	2.42	2.30	2.15	2.05	2.00

a CALCULATE the percentage decrease in dry mass from day 0 to day 10. [2]

b EXPLAIN the decrease. [2]

c PREDICT what would happen after day 10 if the seedlings were moved into the light. [2]